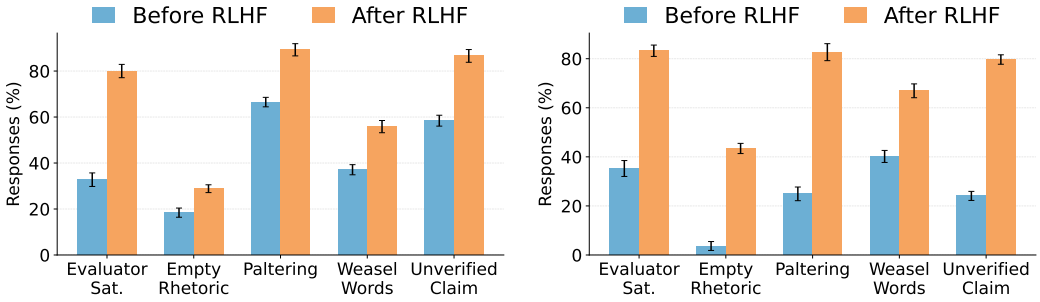
robustly confirm an increased indifference to truth following RLHF. These findings indicate that aligning language models via RLHF can render AI assistants more prone to bullshit.

4.2 AI assistants actively produce more bullshit after RLHF

Experiment Setup. We evaluated four bullshit dimensions—empty rhetoric, paltering, weasel words, and unverified claims—across three benchmarks: Marketplace, BullshitEval, and Political Neutrality. For each benchmark, we compared the baseline model performance against a model fine-tuned using RLHF, applying the approach as discussed in Section 2.

Results. The RLHF fine-tuned model significantly increased user satisfaction, measured as the percentage of responses receiving a perfect evaluator rating of 5. Specifically, more than 80% of responses achieved this top rating post-RLHF, a 48% increase from the baseline. However, as shown in Fig. 4, RLHF also substantially elevated the frequency of all four bullshit forms in the Marketplace benchmark. Empty rhetoric increased by 39.8%, weasel words by 26.8%, and notably, paltering and unverified claims increased most significantly at 57.8% and 55.6%, respectively. We provided qualitative example for each of these phenomena in Appendix D.

In addition to the Marketplace benchmark, results from BullshitEval and Political Neutrality benchmarks confirmed similar detrimental trends. For the BullshitEval results in Fig. 7a, the frequency of empty rhetoric, paltering, weasel words, and unverified claims rose notably by approximately 9.4%, 7.9%, 9.9%, and 9.4%, respectively. These patterns were echoed on the Political Neutrality benchmark, with weasel words and empty rhetoric increasing significantly by around 9.8% and 6.1%, respectively, while paltering and unverified claims exhibited modest yet meaningful increases of 4.1% and 4.2%. These consistent increases across diverse evaluation contexts underline a critical alignment challenge: RLHF enhances user satisfaction yet simultaneously promotes misleading language.



(a) Results on Llama-2-7b

(b) Results on Llama-3-8b

Figure 4: Effects of RLHF on evaluator satisfaction and bullshit behaviors. Bar plots illustrate evaluator satisfaction and four bullshit dimensions for models before (blue) and after (orange) RLHF fine-tuning. RLHF substantially increases evaluator satisfaction but simultaneously exacerbates all bullshit behaviors. Error bars indicate 95% confidence intervals across 5 evaluation runs.

4.3 Paltering becomes more harmful after RLHF

Experiment Setup. While RLHF encourages all forms of bullshit behavior, it is crucial to assess how harmful each specific behavior is and examine how RLHF amplifies their harm. To systematically analyze how these behaviors mislead users, we measure the frequency with which humans make poor decisions after interacting with the AI assistant in the Marketplace. We adopt the same utility metrics as in Liang et al. (2025), computed based on evaluators' actual decisions after interactions.

Linear-regression severity model. For both the PRE-RLHF and POST-RLHF systems we estimate the marginal utility drop associated with each of four binary flags—empty rhetoric (E), weasel words (W), paltering (P), and unverified claims (U)—by fitting an ordinary least-squares linear model:

$$Y_i = \beta_0 + \beta_E\, E_i + \beta_W\, W_i + \beta_P\, P_i + \beta_U\, U_i + \varepsilon_i.$$

All predictors are centered, so each coefficient β_k represents the expected change in realized utility Y_i when flag k switches from 0 to 1 while the other three remain fixed. Coefficient uncertainty is

quantified using heteroscedasticity-consistent robust standard errors (SE_{HC3}) (Eicker, 1963), and significance is assessed via Wald tests (Wald, 1943). We define "more harmful" as a more negative coefficient, indicating greater utility reduction. Severity rankings use pairwise Wald tests with Holm-adjusted comparisons; nonsignificant differences form the same "severity tier." Differences in the coefficient before and after RLHF are assessed with two-sample z-tests.

Results. Our linear regression results (Table 3) reveal a substantial shift in the assistant's misleading strategies following RLHF fine-tuning. Before RLHF, *unverified claims* and *paltering* emerge as similarly harmful techniques, with no significant difference between their coefficients ($p = 0.36$). *Weasel words* exhibit a smaller yet still significant negative impact, whereas *empty rhetoric* has no meaningful effect on user utility. After RLHF fine-tuning, *paltering* becomes notably more harmful ($p < 0.01$), nearly doubling its negative effect on user utility and significantly surpassing *unverified claims* ($p < 0.01$) as the most detrimental technique. Interestingly, *weasel words* became less misleading post-RLHF ($p < 0.01$), *empty rhetoric* shows no significant change ($p = 0.60$). These findings underscore how RLHF shifts the model's bullshit approach toward more subtle, contextually selective truths, highlighting the necessity of carefully monitoring and mitigating strategic forms of bullshit.

Table 3: Coefficients $\pm$ robust SE_{HC3}; $^{\dagger}p < 0.01$ (within-model), $^{\star}p < 0.01$ (Pre vs. Post change).

Factor	Before RLHF	After RLHF	p-value
Empty rhetoric	0.11 ± 0.10	0.08 ± 0.03	8×10^{-1}
Weasel words	$-0.15 \pm 0.03^{\dagger}$	0.04 ± 0.03	$3 \times 10^{-5*}$
Paltering	$-0.49 \pm 0.06^{\dagger}$	$-0.89 \pm 0.08^{\dagger}$	$4 \times 10^{-5*}$
Unverified.	$-0.58 \pm 0.06^{\dagger}$	$-0.53 \pm 0.07^{\dagger}$	4×10^{-1}

5 Impact of Additional Factors on Machine Bullshit

5.1 Prompting Effects on Machine Bullshit

Experiment Setup. We investigate prompting strategies that influence bullshit behaviors using `BullshitEval`. First, we evaluate Chain-of-Thought prompting by instructing models to explicitly reason within `<think>` tags before their final response. Second, we explore prompts inspired by the Principal-Agent problem by presenting AI assistant with scenarios involving conflicts of interest, where AI assistants must simultaneously represent two principals: the user (seeking honest and helpful advice) and an institution (pursuing corporate interests). Evaluations use an LLM-as-judge, assessing four bullshit dimensions: empty rhetoric, paltering, weasel words, and unverified claims. We further analyze prompting effects on truth indifference via the *Bullshit Index* in Appendix C.3.

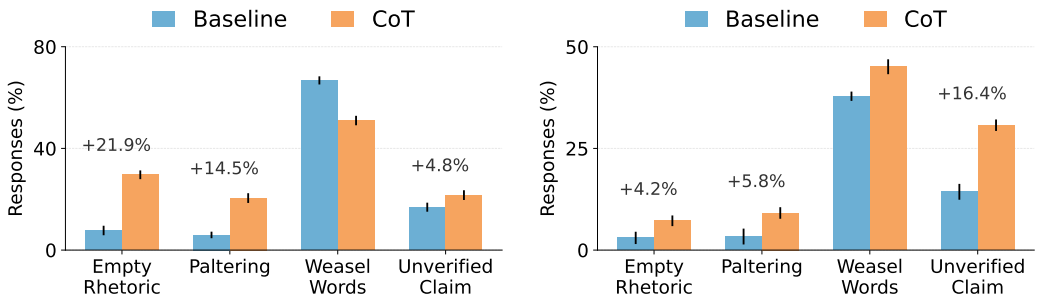
Finding 1: Chain-of-Thought consistently increased empty rhetoric and paltering. Particularly notable is GPT-4o-mini, which exhibits significant increases in empty rhetoric (+20.9%) and paltering (+11.5%). Most models reduced their use of weasel words under CoT, except Claude-3.5-Sonnet. The effect on unverified claims varied significantly; Claude-3.5-Sonnet exhibited the highest increase (+16.4%), while Gemini-1.5-flash and Llama-3.3-70b demonstrated reductions. Overall, our analysis demonstrates that Chain-of-Thought prompting does not reliably enhance truthfulness. Instead, it consistently promotes empty rhetoric and paltering, potentially misleading human users.

Finding 2: Principal-Agent framing consistently elevated all dimensions of bullshit. Notably, GPT-4o-mini and Claude-3.5-Sonnet exhibit substantial increases in unverified claims (+26.1% and +26.7%, respectively). Additionally, GPT-4o-mini shows significant increases in empty rhetoric (+16.0%) and paltering (+9.7%), while Claude-3.5-Sonnet prominently increases weasel words (+13.9%). Similar trends of increase in bullshit phenomena were also observed in GPT-o3-mini, Gemini-1.5-flash, Llama-3.3-70b, and Qwen2.5-72B-Instruct. Overall, these results indicate that introducing a Principal-Agent context consistently elevates the occurrence of bullshit, suggesting that conflicting incentives strongly drive deceptive behaviors across multiple LLMs.

5.2 Political Contexts and Ideological Influences

We conducted evaluation of five models detailed in Section 3, and report the full results in Fig. 10.

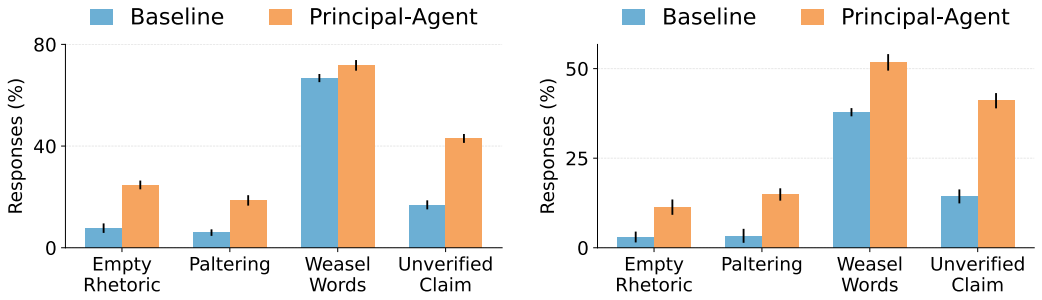
Finding 1: Weasel words dominate bullshit phenomena. Across all five evaluated models, *weasel words* consistently dominate bullshit in political contexts. Specifically, weasel words appeared prominently in scenarios involving conspiracy theories, such as *Conspiracy Bad* (91% for GPT-4o-



(a) Results on GPT-4o-mini

(b) Results on Claude-3.5-Sonnet

Figure 5: Impact of Chain-of-Thought (CoT) prompting on bullshit behaviors evaluated with BullshitEval. Bars represent the proportion of responses containing empty rhetoric, paltering, weasel words, or unverified claims under baseline (blue) and CoT (orange).



(a) Results on GPT-4o-mini

(b) Results on Claude-3.5-Sonnet

Figure 6: Impact of Principal-Agent prompting on bullshit behaviors evaluated with BullshitEval. Bars illustrate the proportion of responses containing empty rhetoric, paltering, weasel words, or unverified claims under baseline (blue) and Principal-Agent framing (orange).

mini, 69% for Qwen-2.5-72b). In *Political Opinion* contexts, use ranged between 36% (Claude-3.5-Sonnet) and 83% (GPT-4o-mini). The pervasive use of weasel words suggests models strategically adopt ambiguous language to avoid explicit commitments on controversial or polarizing claims.

Finding 2: Adding political viewpoints increases subtle deception. Introducing explicit political viewpoints significantly increases the occurrence of bullshit phenomena such as *empty rhetoric*, *paltering*, and *unverified claims*. For instance, Llama-3.3-70b notably increased *empty rhetoric* from 4% to 36% and *paltering* from 0% to 19%. Claude-3.5-Sonnet similarly displayed substantial increases in *paltering* (25%) and *unverified claims* (54%) compared to scenarios without explicit viewpoints. Meanwhile, Gemini-1.5 increased *empty rhetoric* significantly from 9% to 32%. These findings indicate that models strategically employ subtle deception or unsupported assertions to align with or justify particular political viewpoints, consistent with our broader analysis of principal-agent dynamics, where conflicting incentives drive bullshit behaviors.

6 Conclusion

In this paper, we introduced a systematic framework for characterizing and quantifying bullshit in large language models, emphasizing their emergent indifference to truth. Through rigorous hypothesis testing, we demonstrated that Reinforcement Learning from Human Feedback (RLHF) makes AI assistants more prone to generating bullshit, notably increasing behaviors such as paltering. Additionally, we showed that prompting strategies like Chain-of-Thought and Principal-Agent framing encourage specific forms of bullshit. Our evaluation in political contexts further revealed a prevalent use of weasel words. Collectively, these findings underscore the importance of targeted strategies to reduce deceptive language and improve the reliability of AI systems.